

CSE518: User Study Report: Evaluation of Menu Navigation Performance

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1 Introduction

This report details the statistical analysis of two user study scenarios designed to evaluate the performance of four different menu types: *Toolglass*, *Flowmenu*, *Toolpalette*, and *Controlmenu*. The performance metric recorded was task completion time (in seconds).

2 Scenario 1: Between-Subjects Design

In this scenario, 40 participants were recruited, with 10 distinct users assigned to each of the four menu types ($N=40$). This represents a one-way between-subjects design.

2.1 Statistical Analysis

A one-way ANOVA was conducted to determine if there were statistically significant differences in task completion times between the groups.

Table 1: One-way ANOVA Results (Scenario 1)

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Menu	3	0.8998	0.29992	3.455	0.0264
Residuals	36	3.1252	0.08681		

The ANOVA revealed a significant main effect of menu type on completion time ($F(3, 36) = 3.45, p < 0.05$). To identify which specific menus differed, independent pairwise t-tests (without adjustment) were performed.

Key Findings:

- **Toolpalette vs. Toolglass:** Significant difference ($p = 0.0033$).
- **Toolpalette vs. Controlmenu:** Significant difference ($p = 0.0087$).
- Other pairs did not show statistically significant differences.

Table 2: Pairwise t-tests

	Controlmenu	Flowmenu	Toolglass
Flowmenu	0.2053	-	-
Toolglass	0.6769	0.2853	-
Toolpalette	0.0087	0.1856	0.0033

2.2 Visual Analysis

Figure 1 illustrates the distribution of completion times across the four groups.

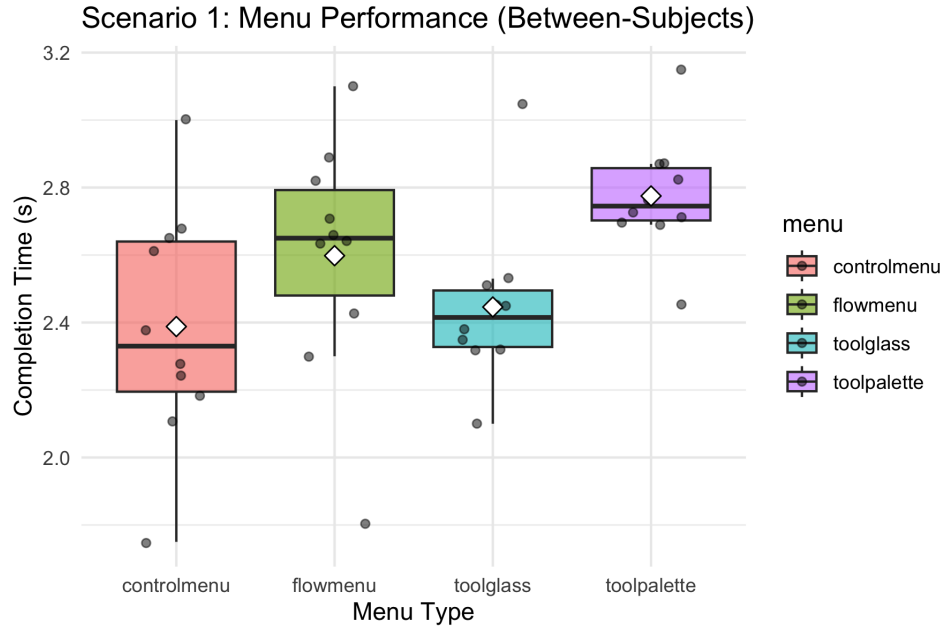


Figure 1: Boxplot of Task Completion Time by Menu Type (Between-Subjects). The diamond shape represents the mean.

Insights: The boxplot shows that the *Toolpalette* menu generally resulted in higher task completion times compared to the *Controlmenu* and *Toolglass*. The variability (interquartile range) is relatively consistent across groups, though outliers are visible in the *Flowmenu* group.

3 Scenario 2: Within-Subjects Design

In this scenario, 10 users tested all four menu types ($N=10$). This represents a repeated-measures (within-subjects) design.

3.1 Statistical Analysis

A Repeated Measures ANOVA was conducted, accounting for the correlation of measurements within the same subject.

Table 3: Repeated Measures ANOVA Results (Scenario 2)

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Menu	3	0.7019	0.23396	10.71	8.19e-05
Residuals	27	0.5898	0.02185		

The analysis showed a highly significant effect of menu type ($F(3, 27) = 10.71, p < 0.001$). Paired t-tests were used for post-hoc analysis.

Table 4: Pairwise t-tests

	Controlmenu	Flowmenu	Toolglass
Flowmenu	0.84598	-	-
Toolglass	0.27632	0.07667	-
Toolpalette	0.00051	0.00166	0.00044

Key Findings:

- **Toolpalette** was significantly slower than all other menus ($p < 0.01$ for all pairs).
- **Flowmenu vs. Toolglass**: Marginally significant ($p = 0.07667$).
- **Flowmenu vs. Controlmenu**: No significant difference ($p = 0.8498$).

3.2 Visual Analysis

Figure 2 displays the distribution of times for the within-subjects trial.

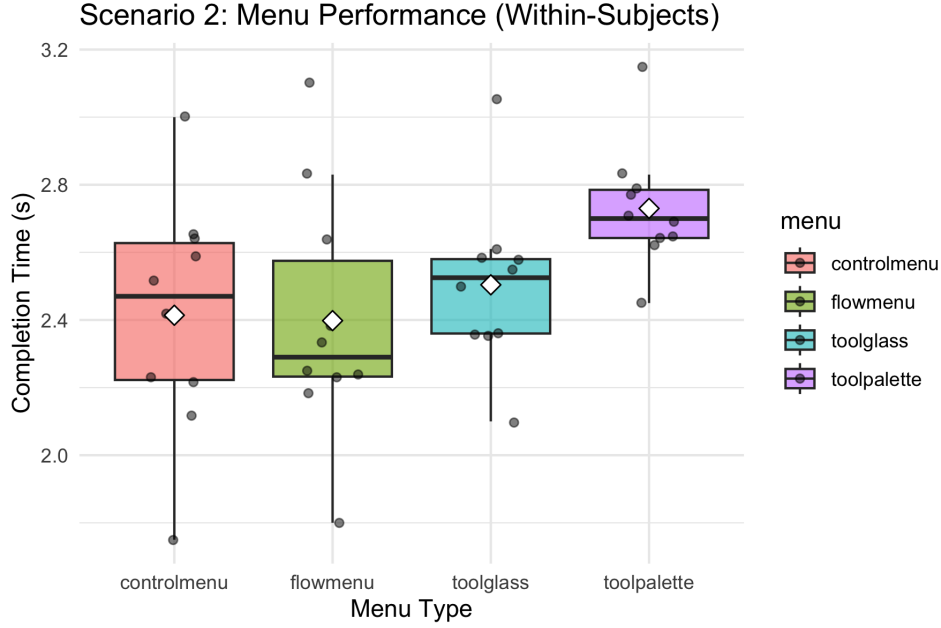


Figure 2: Boxplot of Task Completion Time by Menu Type (Within-Subjects). The diamond shape represents the mean.

Insights: The visualization confirms that the *Toolpalette* consistently underperformed compared to the other techniques. Because this was a within-subjects design, the error variance was reduced, making the differences between the menus more statistically distinct compared to Scenario 1.

4 Conclusion

Both studies consistently identified the *Toolpalette* as the least efficient navigation technique among those tested. However, the **Within-Subjects** design (Scenario 2) provided greater statistical power, revealing highly significant differences ($p < 0.001$) that were less pronounced in the Between-Subjects study ($p \approx 0.026$). Based on these findings, *Control-menu* and *Flowmenu* are the recommended designs for minimizing user task completion time.